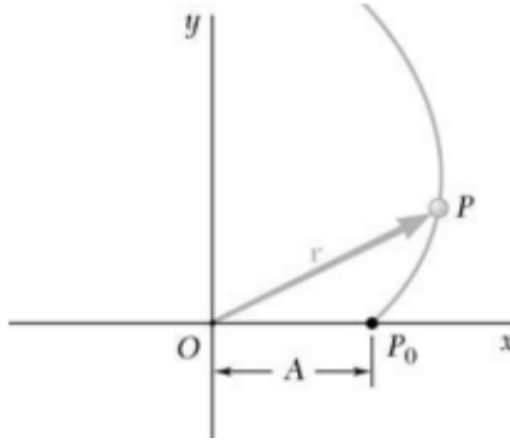

Tutorial 7 Solution sheet
Engineering Mechanics (ME-101)
Date: 11/03/2024 and Time: 7.55 to 8.50 AM

Question 1: The motion of a particle is defined by the position vector $\mathbf{r} = A(\cos t + t \sin t)\mathbf{i} + A(\sin t - t \cos t)\mathbf{j}$, where t is expressed in seconds. Determine the values of t for which the position vector and the acceleration are (a) perpendicular; (b) parallel. **10 Marks**



Solution 1: We have

$$\mathbf{r} = A(\cos t + t \sin t)\mathbf{i} + A(\sin t - t \cos t)\mathbf{j}$$

Then $\mathbf{v} = \frac{d\mathbf{r}}{dt} = A(-\sin t + \sin t + t \cos t)\mathbf{i} + A(\cos t - \cos t + t \sin t)\mathbf{j}$
 $= A(t \cos t)\mathbf{i} + A(t \sin t)\mathbf{j}$ **1 Mark**

and $\mathbf{a} = \frac{d\mathbf{v}}{dt} = A(\cos t - t \sin t)\mathbf{i} + A(\sin t + t \cos t)\mathbf{j}$ **1 Mark**

(a) Where $\mathbf{r}$ and $\mathbf{a}$ are perpendicular, $\mathbf{r} \cdot \mathbf{a} = 0$ **2 Marks**

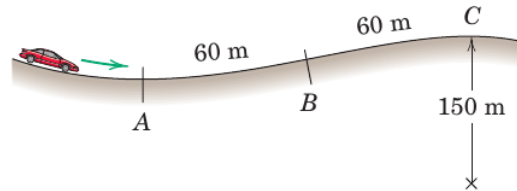
$$\begin{aligned} A[(\cos t + t \sin t)\mathbf{i} + (\sin t - t \cos t)\mathbf{j}] \cdot A[(\cos t - t \sin t)\mathbf{i} + (\sin t + t \cos t)\mathbf{j}] &= 0 \\ \Rightarrow (\cos^2 t - t^2 \sin^2 t) + (\sin^2 t - t^2 \cos^2 t) &= 0 \\ \Rightarrow 1 - t^2 &= 0 \quad \text{or} \quad t = 1 \text{ sec.} \end{aligned}$$
 2 Marks

(b) When $\mathbf{r}$ and $\mathbf{a}$ are parallel, $\mathbf{r} \times \mathbf{a} = 0$. **2 Marks**

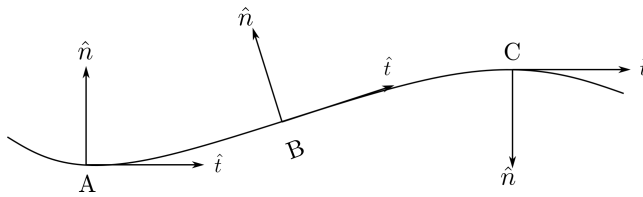
$$\begin{aligned} A[(\cos t + t \sin t)\mathbf{i} + (\sin t - t \cos t)\mathbf{j}] \times A[(\cos t - t \sin t)\mathbf{i} + (\sin t + t \cos t)\mathbf{j}] &= 0 \\ \Rightarrow (\sin t \cos t + t + t^2 \sin t \cos t) - (\sin t \cos t - t + t^2 \sin t \cos t) &= 0 \\ 2t &= 0 \quad \text{or} \quad t = 0. \end{aligned}$$
 2 Marks

Question 2: To anticipate the dip and hump in the road, the driver of a car applies her brakes to produce a uniform deceleration. Her speed is 100 km/h at the bottom A of the dip and 50 km/h at the top C of the hump, which is 120 m along the road from A . If the passengers experience a total acceleration of 3 m/s^2 at A and if the radius of curvature of the hump at C is 150 m, calculate (a) the radius of curvature at A , (b) the acceleration at the inflection point B , and (c) the total acceleration at C .

10 Marks



Solution 2:



In n-t frame,

$$v_A = 100 \text{ km/h} = 27.78 \text{ m/s}$$

$$v_C = 50 \text{ km/h} = 13.89 \text{ m/s}$$

$$S = 120 \text{ m}$$

$$|\bar{a}_A| = 3 \text{ m/s}^2, \quad \rho_C = 150 \text{ m}$$

Uniform deceleration along the curve,

$$\text{We have,} \quad v_C^2 = v_A^2 - 2a_t S \Rightarrow a_t = \frac{v_A^2 - v_C^2}{2 \times S} \Rightarrow a_t = 2.412 \text{ m/s}^2 \quad \text{1 Mark}$$

$$|\bar{a}_A| = (a_t^2 + a_n^2)^{1/2} = 3 \quad \text{1 Mark}$$

$$a_n = (a_A^2 - a_t^2)^{1/2} \Rightarrow a_n = 1.78 \text{ m/s}^2$$

$$(a) \quad (a_n)_A = \frac{v_A^2}{\rho_A} \Rightarrow \rho_A = 432.61 \text{ m} \quad \text{2 Marks}$$

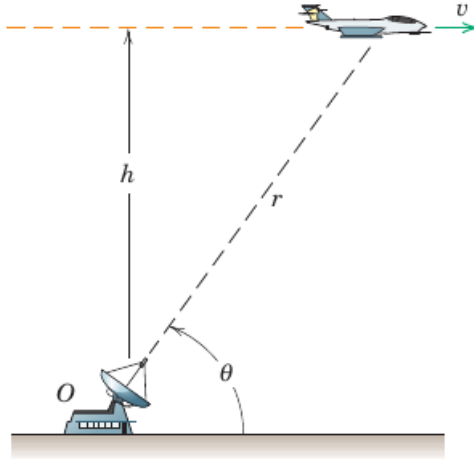
$$(b) \quad (a)_B = \cancel{(a_n)_B}^0 + (a_t)_B \Rightarrow (a)_B = (a_t)_B = 2.41 \text{ m/s}^2 \quad \text{3 Marks}$$

$$\begin{aligned} (c) \quad (\bar{a})_C &= \frac{v_C^2}{\rho_c} \hat{n} + (a_t)_A \hat{t} \\ &= 1.28 \hat{n} + 2.41 \hat{t} \\ ||a||_C &= \sqrt{1.28^2 + 2.41^2} = 2.73 \text{ m/s}^2 \quad \text{3 Marks} \end{aligned}$$

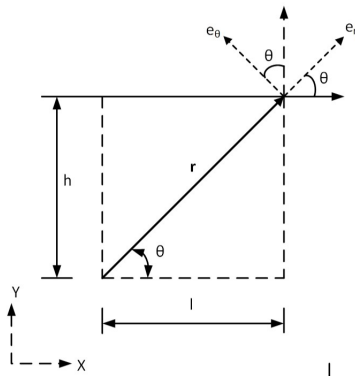
Question 3: A jet plane flying at a constant speed v at an altitude $h = 10 \text{ km}$ is being tracked by radar located at O directly below the line of flight. If the angle θ is decreasing at the rate of 0.020 rad/s when $\theta = 60^\circ$, determine the value of $\ddot{r}$ at this instant and the magnitude of the

velocity $\mathbf{v}$ of the plane.

10 Marks



Solution 3:



Polar coordinate system,

$$v = \text{constant}$$

$$\bar{\mathbf{v}} = \dot{r}\hat{\mathbf{e}}_r + r\dot{\theta}\hat{\mathbf{e}}_\theta$$

$$h = l \tan 60^\circ \implies l = h/\sqrt{3}$$

$$r = (h^2 + h^2/3)^{1/2} \implies r = 2h/\sqrt{3} \quad \text{1 Mark}$$

As the motion is completely horizontal,

$$\dot{\mathbf{r}} = \dot{r}\hat{\mathbf{e}}_r + r\dot{\theta}\hat{\mathbf{e}}_\theta$$

$$\hat{\mathbf{e}}_r = (\cos \theta \hat{\mathbf{i}} + \sin \theta \hat{\mathbf{j}}), \quad \hat{\mathbf{e}}_\theta = (-\sin \theta \hat{\mathbf{i}} + \cos \theta \hat{\mathbf{j}})$$

$$\implies \bar{\mathbf{v}} = (\dot{r} \cos \theta - r\dot{\theta} \sin \theta)\hat{\mathbf{i}} + (\dot{r} \sin \theta + r\dot{\theta} \cos \theta)\hat{\mathbf{j}} \quad \text{1 Mark}$$

Now,

$$v_y = 0,$$

$$\dot{r} = -r\dot{\theta}(\cot \theta) \implies \dot{r} = -\frac{2h}{\sqrt{3}}\dot{\theta}\frac{1}{\sqrt{3}} = -\frac{2}{3}h\dot{\theta}$$

1 Mark

$$\bar{\mathbf{a}} = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{e}}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\mathbf{e}}_\theta = (a_r\hat{\mathbf{e}}_r + a_\theta\hat{\mathbf{e}}_\theta)$$

$$\bar{\mathbf{a}} = (a_r \cos \theta - a_\theta \sin \theta)\hat{\mathbf{i}} + (a_r \sin \theta + a_\theta \cos \theta)\hat{\mathbf{j}}$$

Now,

$$a_x = 0 \implies a_r \cos \theta = a_\theta \sin \theta$$

Assuming $\dot{\theta}$ is constant i.e.,

$$\ddot{\theta} = 0,$$

$$\Rightarrow \ddot{r} = (r\dot{\theta}^2 + 2\dot{r}\dot{\theta}\tan\theta)$$

1 Mark

Calculations,

$$\dot{r} = -\frac{2}{3} \times 10 \times (-0.02) \times 3600 = 479.88 \text{ km/h}$$

1 Mark

$$r\dot{\theta} = \frac{2}{\sqrt{3}} \times 10 \times (-0.02) \times 3600 = -828 \text{ km/h}$$

1 Mark

$$|v| = \left(\frac{4}{9} \times 100 + \frac{4}{3} \times 100\right)^{1/2} \times 0.02 \times 3600 \text{ km/h}$$

$$= 960 \text{ km/h}$$

2 Marks

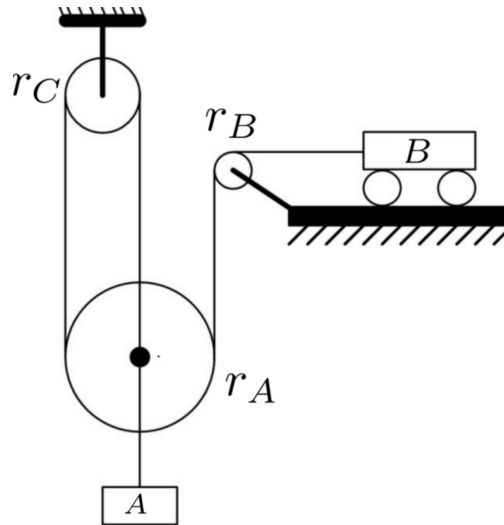
$$\ddot{r} = \left(\frac{20}{\sqrt{3}} \times 0.02^2 - 2 \times \frac{20}{3} \times 0.02^2 \times \tan 60^\circ\right)$$

$$\Rightarrow \ddot{r} = -4.6188 \text{ m/s}^2$$

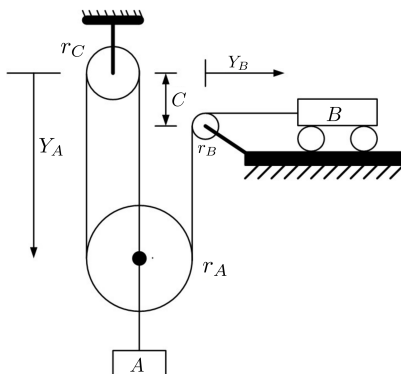
2 Marks

Question 4: If the block B has a leftward velocity of 1.2 m/s, determine the velocity of block A .

10 Marks



Solution 4:



$$L = 2Y_A + (Y_A - c) + \pi r_A + \pi r_C + \frac{\pi}{2}r_B + Y_B$$

$$= \text{constant}$$

5 Marks

$$\dot{L} = 0 = 3\dot{Y}_A + \dot{Y}_B$$

2 Marks

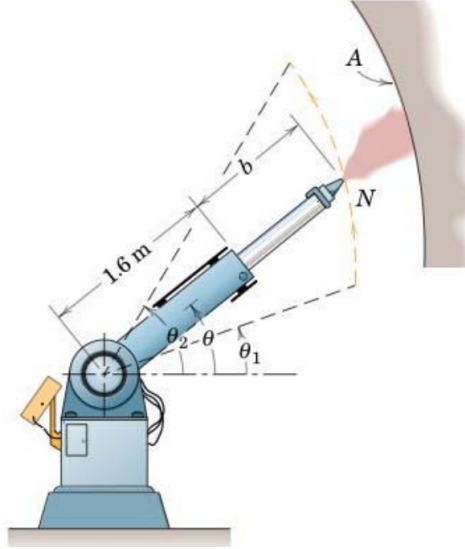
$$\Rightarrow \dot{Y}_A = -\frac{1}{3}\dot{Y}_B = -\frac{1}{3} \times (-1.2)$$

$$\Rightarrow \dot{Y}_A = 0.4 \text{ m/s}$$

3 Marks

Question 5: The paint spraying robot is programmed to paint a production line of curved surface A (seen on edge). The length of the telescoping arm is controlled according to $b = 0.3 \sin(\pi t/2)$ where b is in meter and t is in seconds. Simultaneously, the arm is programmed to rotate according to $\theta = \frac{\pi}{4} + \frac{\pi}{8} \sin(\pi t/2)$ rad. Calculate the magnitude of the velocity and acceleration of the nozzle N at $t = 1$ sec.

10 Marks



Solution 5:

$$r = 1.6 + 0.3 \sin\left(\frac{\pi t}{2}\right) \text{ m}; \quad \dot{r} = 0.3 \left(\frac{\pi}{2}\right) \cos\left(\frac{\pi t}{2}\right) \text{ m/s}; \quad \ddot{r} = -0.3 \left(\frac{\pi}{2}\right)^2 \sin\left(\frac{\pi t}{2}\right) \text{ m/s}^2 \quad \text{1 Mark}$$

$$\theta = \frac{\pi}{4} + \frac{\pi}{8} \sin\left(\frac{\pi t}{2}\right); \quad \dot{\theta} = \frac{\pi}{8} \times \frac{\pi}{2} \cos\left(\frac{\pi t}{2}\right) = \frac{\pi^2}{16} \cos\left(\frac{\pi t}{2}\right) \text{ rad/s}$$

$$\ddot{\theta} = -\frac{\pi^2}{16} \times \frac{\pi}{2} \sin\left(\frac{\pi t}{2}\right) = -\frac{\pi^3}{32} \sin\left(\frac{\pi t}{2}\right) \text{ rad/s}^2 \quad \text{1 Mark}$$

$$\text{Velocity in the radial direction } v_r = \dot{r} = \frac{0.3\pi}{2} \cos\left(\frac{\pi t}{2}\right) \text{ m/s.} \quad \text{1 Mark}$$

$$\text{Velocity in } \theta \text{ direction } v_\theta = r\dot{\theta} = \left(1.6 + 0.3 \sin\left(\frac{\pi t}{2}\right)\right) \frac{\pi^2}{16} \cos\left(\frac{\pi t}{2}\right) \text{ rad/s.} \quad \text{1 Mark}$$

$$\text{Acceleration in radial direction } a_r = \ddot{r} - r\dot{\theta}^2$$

$$= -0.3 \left(\frac{\pi}{2}\right)^2 \sin\left(\frac{\pi t}{2}\right) - \left(1.6 + 0.3 \sin\left(\frac{\pi t}{2}\right)\right) \left(\frac{\pi^2}{16} \cos\left(\frac{\pi t}{2}\right)\right)^2 \quad \text{1 Mark}$$

$$\text{Acceleration in } \theta \text{ direction } a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

$$= \left(1.6 + 0.3 \sin\left(\frac{\pi t}{2}\right)\right) \left(-\frac{\pi^3}{32} \sin\left(\frac{\pi t}{2}\right)\right) + 2 \left(0.3 \left(\frac{\pi}{2}\right) \cos\left(\frac{\pi t}{2}\right)\right) \left(\frac{\pi^2}{16} \cos\left(\frac{\pi t}{2}\right)\right) \quad \text{1 Mark}$$

At time $t = 1\text{sec}$

$$\text{Velocity in radial direction } v_r = \dot{r} = \frac{0.3\pi}{2} \cos\left(\frac{\pi}{2}\right) = 0$$

$$\text{Velocity in } \theta \text{ direction } v_\theta = r\dot{\theta} = \left(1.6 + 0.3 \sin\left(\frac{\pi}{2}\right)\right) \frac{\pi^2}{16} \cos\left(\frac{\pi}{2}\right) = 0$$

$$\text{Velocity } v = \sqrt{v_r^2 + v_\theta^2} = 0$$

2 Marks

$$\text{Acceleration in the radial direction } a_r =$$

$$= -0.3\left(\frac{\pi}{2}\right)^2 \sin\left(\frac{\pi}{2}\right) - \left(1.6 + 0.3 \sin\left(\frac{\pi}{2}\right)\right) \left(\frac{\pi^2}{16} \cos\left(\frac{\pi}{2}\right)\right)^2 = -0.74 \text{ m/s}^2$$

$$\text{Acceleration in } \theta \text{ direction } a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

$$= \left(1.6 + 0.3 \sin\left(\frac{\pi}{2}\right)\right) \left(-\frac{\pi^3}{32} \sin\left(\frac{\pi}{2}\right)\right) + 2\left(0.3\left(\frac{\pi}{2}\right) \cos\left(\frac{\pi}{2}\right)\right) \left(\frac{\pi^2}{16} \cos\left(\frac{\pi}{2}\right)\right)$$

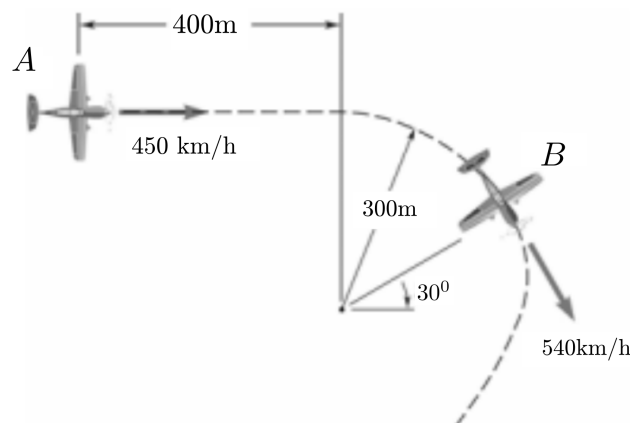
$$= -1.841 \text{ m/s}^2$$

$$\text{Acceleration } a = \sqrt{a_r^2 + a_\theta^2} = 1.984 \text{ m/s}^2$$

2 Marks

Question 6: At a given instant in an airplane race, airplane A is flying horizontally in a straight line, and its speed is being increased at the rate of 8 m/s^2 . Airplane B is flying at the same altitude as airplane A and, as it rounds a pylon, is following a circular path of 300 m radius. Knowing that the given instant the speed of B is being decreased at the rate of 3 m/s^2 , determine, for the positions shown, (a) the velocity of B relative to A , (b) the acceleration of B relative to A .

10 Marks



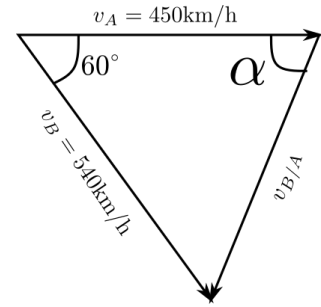
Solution 6:

(a)

$$v_A = 450 \text{ km/h} \quad v_B = 540 \text{ km/h} = 150 \text{ m/s}$$

We have $v_B = v_A + v_{B/A}$ **2 Marks**

The graphical representation of this equation is then as shown.



We have $\Rightarrow v_{B/A}^2 = 450^2 + 540^2 - 2(450)(540) \cos 60^\circ \Rightarrow v_{B/A} = 501.10 \text{ km/h}$

and $\frac{540}{\sin \alpha} = \frac{501.10}{\sin 60^\circ} \Rightarrow \alpha = 68.9^\circ$

$\Rightarrow v_{B/A} = 501 \text{ km/h}$ (68.9° from -x axis anticlockwise) **3 Marks**

(b)

$$a_A = 8 \text{ m/s}^2 \quad (a_B)_t = 3 \text{ m/s}^2 \quad (60^\circ \text{ from the -x axis clockwise})$$

Now $(a_B)_n = \frac{v_B^2}{\rho_B} = \frac{(150 \text{ m/s})^2}{300 \text{ m}} \Rightarrow (a_B)_n = 75 \text{ m/s}^2$ (30° from -x axis anticlockwise)

Then $a_B = (a_B)_t + (a_B)_n$ **2 Marks**

$$= 3(-\cos 60^\circ \mathbf{i} + \sin 60^\circ \mathbf{j}) + 75(-\cos 30^\circ \mathbf{i} - \sin 30^\circ \mathbf{j})$$

$$= -(66.452 \text{ m/s}^2) \mathbf{i} - (34.902 \text{ m/s}^2) \mathbf{j}$$

Finally $a_B = a_A + a_{B/A}$

$$a_{B/A} = (-66.452 \mathbf{i} - 34.902 \mathbf{j}) - (8 \mathbf{i})$$

$$= -(74.452 \text{ m/s}^2) \mathbf{i} - (34.902 \text{ m/s}^2) \mathbf{j}$$

$a_{B/A} = 82.2 \text{ m/s}^2$ (25.1° from the -x axis anticlockwise) **3 Marks**